

MAYAN NOMENCLATURE

BY
CHARLES P. BOWDITCH

MAYAN NOMENCLATURE

BY
CHARLES P. BOWDITCH

The Project Gutenberg eBook of Mayan Nomenclature

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Mayan Nomenclature

Author: Charles P. Bowditch

Release date: April 20, 2022 [eBook #67879]

Most recently updated: October 18, 2024

Language: English

Original publication: United Kingdom: Cambridge The University Press,
1906

Other information and formats: www.gutenberg.org/ebooks/67879

Credits: The Online Distributed Proofreading Team at <https://www.pgdp.net>
(This file was produced from images generously made available
by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK MAYAN
NOMENCLATURE ***

MAYAN NOMENCLATURE

BY
CHARLES P. BOWDITCH

Privately Printed

CAMBRIDGE
THE UNIVERSITY PRESS
1906

It has long been well known that the Mexican numeration is vigesimal, and, as far as I know, there is no proof that it was ever used in the calculation of long reaches of time. The Cakchiquel numeration is also vigesimal, and Brinton states (Maya Chronicles, p. 44) that the Maya numeration is also vigesimal, giving

20 units	=	one kal	=	20
20 kal	=	one bak	=	400
20 bak	=	one pic	=	8000
20 pic	=	one calab	=	160000, etc.

But this Maya system is never used in connection with days. Wherever a long number of days is referred to, we find invariably the following system in use:

20 units	=	one of the second term
18 of the second term	=	one of the third term
20 of the third term	=	one of the fourth term
20 of the fourth term	=	one of the fifth term.

And this system is, as far as we know, used for nothing else but for reckoning days. The only difference between the first system and the second is that in the first system twenty of the second term equal one of the third, while in the second system eighteen of the second equal one of the third. This difference is, of course, essential, and it seems most probable that the change was introduced in order to bring the third term as near the length of the year as possible, and to conform the day numeration to the number of days and months in the year.

Professor Cyrus Thomas is unwilling to see in this anything more than the counters by which to count the days, and denies to it the name of a calendar; but as the system of day numeration is different from the usual system, and is used only for counting days, and as this system counts forward in almost every case in the inscriptions, and in a majority of cases in the Dresden Codex, from a fixed date, 4 Ahau 8 Cumhu, it seems impossible to see any difference between it and a calendar system.

It was certainly to be hoped that the designations which Dr. Seler gave to these Maya periods of time in his “Die Monumente von Copan, etc.”^[1]

would have been accepted by Americanists, especially since very good reasons were given by Seler for their adoption. But this view does not meet the approval of Dr. Förstemann or Dr. Thomas. The latter in his "Maya Calendar, Part II.," published in the "Report of the Bureau of American Ethnology," still calls the period of 20 days a chuen and that of 360 days an ahau, while the former in his article on "Die Lage der Ahaus bei den Mayas," published in Part I. of the 1904 issue of the "Zeitschrift für Ethnologie," makes the following statement: "The katun has also been supposed to be $24 \times 365 = 8760$ days long (and I held this view for a long time), indeed the long period of $52 \times 365 = 18,980$ days is also occasionally designated with the word, while the sixth multiple of this member or 113,800 is called an ahaukatun." He uses the terms "day," "uinal," and "ahau" for the periods of 1, 20, and 7200 days respectively.

[1] See Seler, "Gesammelte Abhandlungen zur Amerikanischen Sprach- und Alterthumskunde," Vol. I. p. 722.

Such statements from such learned scholars must attract attention, especially as it is not the first time that they have made similar statements. If they give a correct statement of facts, it shows that the system of the Maya numeration or calendar was in a woful condition, as far as its nomenclature was concerned. It will be well, therefore, to take up this question of nomenclature anew. In all matters of this kind it is wise, while giving due value to the views of later writers, to place the most dependence for the solution of such questions upon contemporary or nearly contemporary evidence.

And where is such evidence to be found? Though there are several Spanish writers who lived in the sixteenth century who have written on the Nahuas, Bishop Landa is the earliest Spanish authority who has dealt with Maya customs and history. And the only Maya authorities of an early date that the student has access to are the papers contained in Brinton's Maya Chronicles, the principal of which are the Books of Chilán Balam, which Dr. Brinton declares "to constitute about all that remains to us ... of the ancient history of the peninsula" as far as he knows. Yet, in making objection to certain views of Dr. Seler, Dr. Förstemann says, "It is based on certain statements in the Books of Chilán Balam, a very dubious source according to Seler's own assertion." I am at a loss to know why such sweeping condemnation should be made of these books, for though the

copies from which Dr. Brinton quotes may not be very old, these copies hand down to us records which must be of very great age. Dr. Förstemann himself quotes from them with approval, and it is certain that whatever may be their historical value, the evidence which they give incidentally cannot fail to be of great value. This evidence will, I think, prove—

First, that the period of 20 times 360 days was called a katun and not an ahau.

Second, that each of the constituent parts of a katun was called a tun.

Third, that no such period of time as an ahaukatun is mentioned in the Books of Chilán Balam.

The first assertion may seem difficult to prove when in fact the larger number of cases where a separate katun is mentioned in the Books of Chilán Balam the word “ahau” seems to be substituted for “katun.” This is true in all five of the Books of Chilán Balam published by Brinton. But the reason of this is very evident, for it will be seen by a careful perusal that the word “ahaus” is never used when katuns in general are spoken of, and that whenever the word “ahau” is apparently given as a synonym of “katun,” it is merely the name of a particular katun and it always has a number attached to it and this number is the number of the day ahau with which a given katun ended. Further, this number is never an ordinal number, as translated by Brinton, but a cardinal number. Therefore 1, 2, 3, 4, 5, 6, etc. should be substituted in Brinton’s translation for 1st, 2d, 3d, 4th, 5th, 6th, etc., and in the quotations from Brinton this substitution is made.

It may be here stated that the inscriptions all show that where an even uinal is given (and therefore where an even tun, katun, or cycle is given) the day is Ahau. If then it was desirable to distinguish the katuns from each other, two methods could be used: either count them numerically, 1, 2, 3, 4, etc., or name them from the day Ahau with which the preceding katun had ended. The former method is found in the inscriptions and the Dresden Codex, the latter in the Books of Chilán Balam. The second method would not be possible if each katun ended with the same numbered ahau. But 7200 is not divisible by 13 without a remainder, but equals $13 \times 553 + 11$. If then a particular katun ended with 13 Ahau, the next would end with Ahau, but the number attached to Ahau would be $13 + 11$, or, deducting thirteens, 11. The next katun would end with 9, the next with 7, and so on. The katuns

then would be known as katuns, 13 Ahau, 11 Ahau, 9 Ahau, 7 Ahau, 5 Ahau, 3 Ahau, 1 Ahau, 12 Ahau, 10 Ahau, 8 Ahau, 6 Ahau, 4 Ahau, 2 Ahau, 13 Ahau, etc.

Taking up, therefore, No. I. of the Books of Chilán Balam, published by Brinton—that of Mani—we find in the paragraph numbered 1 by Brinton “This is the arrangement of the katuns” not “of the ahaus,” and in paragraph numbered 2 “Four katuns had passed, etc.” not “four ahaus.” This is followed by the statement “When they set out for this country, it was Ahau 8,” not “the 8th Ahau.” And then follows “6 Ahau, 4 Ahau, 2 Ahau, fourscore years and one year, for it was Tun 1, 13 Ahau when, etc.” That 6 Ahau, 4 Ahau, 2 Ahau refer to the katuns is very clear, and that 4 katuns with the names 8 Ahau, 6 Ahau, 4 Ahau, and 2 Ahau are called fourscore years is equally clear. In paragraph 4 we have “4 Ahau, 2 Ahau, 13 Ahau—threescore years they ruled Ziyan caan, etc.” Here three katuns are called threescore years.

This would seem to show that the katuns called 4 Ahau, 2 Ahau, 13 Ahau, were each equal to a score of years of 365 days each. When, however, we try to account for the numbering of the katuns on this basis, we find that the numbers of the ahaus ending each katun would come in the following order: 11. 5. 12. 6. 13. 7. 1. 8. 2. 9. 3. 10. 4. 11, etc., while the real order is given in the books as 11. 9. 7. 5. 3. 1. 12. 10. 8. 6. 4. 2. 13, etc. If the word “haab” or the Spanish “años,”^[2] which occurs in paragraph 3, is taken literally, there would seem to be no explanation of this difficulty; but if we consider that these books used these words as we often use them now as meaning approximately “years,” and if we substitute the third term of the numeral series as found in the codices for the word “years”—in other words, if we substitute 360 for 365—we find then that the katuns or scores of 360 days will end with a day Ahau with the numbers 11. 9. 7. 5. 3. 1. 12. 10, etc., as has been said and as given in the Books of Chilán Balam. This has been shown by Seler, Goodman, and others.

^[2] It is very probable that “años” is merely a gloss—and an erroneous one at that.

It will then be a good working theory that the score of 360 days is called a katun, and that each katun is distinguished by the name of an ahau with its proper number, with which a katun ended. That the ahau and its number are merely the name of the katun and not the katun itself is shown in No. 5 of

Brinton's Books of Chilán Balam,—the Book of Chumayel. Here in paragraph 1 we have "4 Ahau was the name of the katun," in paragraph 3 "4 Ahau was the name of the katun," and in paragraph 7 "11 Ahau was the name of the katun." To say that "ahau" was a synonym for "katun" would be as foolish as to say of a family containing John Smith, Mary Smith, and James Smith, that John and Mary and James were all synonyms of Smith, when they were merely names to distinguish one Smith from another.

This is also shown in the Book of Mani (pp. 96 et seq. of Brinton's "Maya Chronicles"), where

- Par. 8 says "Lai u katunil cabil ahau," "In the katun 2 Ahau."
- " 9 says "Lai u katunil buluc ahau," "In the katun 11 Ahau."
- " 10 says "Laili u katunil uaxac ahau," "In the katun 8 Ahau."
- " 12 says "lay u katunil uac ahau," "In the katun 6 Ahau."
- " 15 says "u katunil ho ahau," "In the katun 5 Ahau."

Also in the Book of Tizimin (pp. 139 et seq. of Brinton):

- Par. 6, "Uaxac ahau—lay u katunil," "8 Ahau—in this katun."
- " 14, "Bolon ahau—lai—u katunil," "9 Ahau—in this katun."
- " 15, "Vuc ahau—u katunil," "7 Ahau—in this katun."

Also in the first Book of Chumayel (pp. 154 et seq. of Brinton):

- Par. 2, "Uaxac ahau—layli u katunil," "8 Ahau—in this katun."
- " 8, "Bolon ahau—u katunil," "9 Ahau—in this katun."

Also in the second Book of Chumayel (pp. 166 et seq. of Brinton):

- Par. 3, "Uaxac ahau u katunil," "8 Ahau in this katun."
"laix u katunil," "in this katun."
- " 5, "Hun ahau—lay u katunil," "1 Ahau—in this katun."
- " 8, "Buluc ahau—laix u katunil," "11 Ahau—in this katun."
"Buluc ahau. Laix u katunil," "11 Ahau in this katun."
- " 9, "Bolon ahau—lay katun," "9 Ahau—in this katun."
"Uac Ahau—lay u katunil," "6 Ahau—in this katun."

I can find no excuse for considering "ahau" as a synonym for "katun" in these Books of Chilán Balam. And *a priori* it is difficult to conceive of a numeral system being made up by a nation as cultivated as the Mayas, by which the name of a day should be taken to mean a period of time, with all

the chances of error which might arise from such use. Moreover, if this were the case, it would be natural to suppose that in the inscriptions and codices the day sign ahau might be found meaning 7200 days, and yet I am unaware that a single instance of this exists on the monuments or in the codices. There are one or two cases in the inscriptions where this has been suggested as being possible, but it is very far from being proved or from even having strong evidence in favor of such an explanation.

Dr. Förstemann says that at times the name “katun” was given to the period of 18,980 days; but the only authority for this use is, as far as I know, Pio Perez, who says that some applied the name to a “lustre of 4 years,” while others thought that “13 years completed the katun.”^[3] Again Pio Perez speaks of “the cycle of 52 years called by the Indians katun.” In none of these cases is any authority given for this use.^[4]

^[3] Stephens, “Travels in Yucatan,” Appendix, p. 439.

^[4] Ibid., p. 440.

It is probably safe to say that any use of the word “ahau” as meaning a katun, or any use of the word “katun” as meaning anything but 20 times 360 days, or any use of the day symbol ahau as meaning a period of time, does not occur in any evidence which could be called contemporary or even approximately so. Don Pio Perez is, I fear, the only author who can be cited for the other side, and his opinion, though worthy of being listened to, is not to be relied upon without the support of, or against the evidence of, early authorities.

The second question as to the name to be given to the period of 360 days must be decided by the same authority.

We have found that a katun is probably a score of periods of 360 days each, and it would be natural to suppose that the constituent parts of the katun, as mentioned in the Books of Chilán Balam, would be these periods of 360 days, but of this there is no absolute proof in the Books of Chilán Balam. The proof of this, however, is so strong in the Dresden Codex, where the series runs, 20 of the first order equals 1 of the second order, 18 of the second order equals 1 of the third order, and 20 of the third order equals 1 of the fourth, that no time need be wasted upon this part of the question. We do know, however, that each of these constituent parts is called a tun and that they are numbered as high as 13.

The following list gives, I think, all the cases where the word “tun” is used in the Books of Chilán Balam:

Book of Mani, paragraph 2, “Hun piztun oxlahun ahau,” which Brinton translates “the first year of the thirteenth ahau.” I think the proper translation to be “Tun 1 of Ahau 13,” meaning that 1 tun had passed of Katun 13 Ahau.

Paragraph 9, “tu lahun tun uaxac ahau” is translated by Brinton, “and it was the tenth year of the eighth ahau,” while it should be “in Tun 10 of Katun 8 Ahau.”

In the Book of Tizimin we find that in

Paragraph 1, “tu humpiztun ahoxlahunahau” is translated “to the first year of the thirteenth ahau,” while it should be “to (or in) Tun 1 of Katun 13 Ahau.”

Paragraph 11, “Cabil ahau; oxlahun tun” is translated “The second ahau; on the thirteenth, etc.” It should be “on Katun 2 Ahau Tun 13, etc.”

Also in the second Book of Chumayel, we find in

Paragraph 2, “tu yoxpiztun ychil hun ahau,” which is translated “on the third year in the first ahau,” when it should be “in Tun 3 of Katun 1 Ahau.”

Paragraph 3, “Tu uucpiztun uaxac ahau u katunil” is translated “in the seventh year of the eight Ahau katun,” when it should be “in Tun 7 of Katun 8 Ahau.”

Paragraph 8, “tu hunpiztun Buluc ahau. Laix u katunil” is translated “in the first year of the eleventh ahau, it was also in this katun,” when it should be “on Tun 1 of Katun 11 Ahau. In this Katun, etc.” Also “tu uucpiztun Buluc ahau u katunil” is translated “in the seventh year of the eleventh ahau katun,” when it should be “in Tun 7 of Katun 11 Ahau.”

These two katuns are evidently the same—one date being in Tun 1 and the other in Tun 7 of the same katun—and yet Brinton translated the first as being in the 11th katun and the other as being in the 11th ahau katun.

Paragraph 9, “tu uacpiztun ychil ahBolon ahau katun lae.” is translated “in the sixth year of the ninth ahau katun.” It should be “in Tun 6 of Katun 9 Ahau.”

Although Brinton has almost always translated “tun” by “year,” it is very evident from the above that the Mayas called the native division of the katun “tun.”

It may further be noted that the word “kin” is used for day in the way of numeration in the Book of Mani, paragraph 13, “9 Imix was the day on which Ahpula died” and in paragraph 13 of the Book of Tizimin and paragraph 7 of the first Book of Chumayel, the same statement is made.

May it not then be fairly assumed that the Mayas called the day or the unit of their calendrical numeration “kin,” their 360-day period “tun,” and the period of 20 tuns “katun,” as Dr. Seler has suggested?

The third question is as to the so-called period of “ahau katun.” I give below all the cases in which Brinton has used the word, and in all but one of them the Maya has “ahau u katunil”; and in one case Brinton has translated a sentence as “it was the eleventh ahau katun,” while exactly similar phrases (with the exception of the number) have been translated by him as “4 Ahau was the name of the katun.” In the phrase in paragraph 9 of the second Book of Chumayel, where “ah Bolon ahau katun” appears, the whole context makes it evident that a katun and not an ahau katun, if there was such a period, is meant. The paragraph says “9 Ahau; no stone was taken at this time; in this katun came the Bishop Brother Francisco Toral; he arrived in Tun 6 of Katun 9 Ahau”; and yet Brinton makes it read in the first clause that while Bishop Toral came in a katun called 9 Ahau, he “arrived in Tun 6” of the “ninth Ahau katun.”

Dr. Tozzer, whose knowledge of the Maya languages, gained in long residence among the Maya Indians of Yucatan and Chiapas, enables him to speak with authority, writes me as follows: “The form u katunil of ahau ukatunil, for example,” in paragraph 3 of the second Book of Chumayel, p. 167, “is probably possessive, and it would then be literally ‘the Ahau 8, its Katun’ (the Ahau’s Katun) which might strengthen the point against joining the two together as Ahaukatun.”

In the second Book of Chumayel, p. 167, paragraph 3, “uucpiztun uaxac ahau u katunil; laix u katunil, etc.” Brinton translates “in the 7th year of the 8th Ahau katun, in this katun, etc.” There is no authority for translating ahau u katunil “ahau katun.” It should be “In Tun 7 of Katun 8 Ahau, in this

katun, etc.,” or, as suggested by Dr. Tozzer, “In Tun 7 of Ahau eight’s Katun.”

Again, in paragraph 5, the translation of “Hun ahau—tu hunpiztun ychil hun ahau u katunile” is given, “The first ahau—in the first year of the first ahau katun.” It should be “1 Ahau—in Tun 1 of the Katun 1 Ahau, &c.”

Again, in paragraph 8 “tu uucpiztun Buluc Ahau u katunil” is translated “in the seventh year of the eleventh ahau katun,” when it should be “in Tun 7 of Katun 11 Ahau.” Just above “Tun 1 of 11 Ahau” is given and followed by “in this katun.”

Again in paragraph 9 we read “Bolon Ahau—lay katun yax ulci obispo Fray Fran’co to Ral huli tu uacpiztun ychil ahBolon ahau katun lae,” which Brinton translates “The ninth ahau—in this katun first came the bishop Brother Francisco Toral; he arrived in the sixth year of the ninth ahau katun.” It should be “9 Ahau—in this katun—in Tun 6 of Katun 9 Ahau.” This is the only place in the Maya where the words “ahau katun” appear together. But the context makes it clear that “ahau katun” is not a period of time. The paragraph begins with “9 Ahau,” and speaks of it as a katun during which Bishop Toral arrived, and then proceeds to say that he arrived in Tun 6 of this “ahBolon ahau katun,” which can mean nothing else than the “Katun 9 Ahau.”

In the third Book of Chumayel, paragraphs 5 and 6, Brinton translates “Can ahau u katunil” as “The fourth ahau katun” instead of “Katun 4 Ahau.”

In paragraph 7 “Uaxac ahau u katunil” is translated “The eighth ahau katun” instead of “Katun 8 Ahau” (note here that this is the same “8 Ahau,” when Chichen Itza was deserted for Champoton).

Also “Oxlahun ahau u katunil” is translated “In the thirteenth ahau katun” instead of “Katun 13 Ahau.”

Also “Buluc ahau u kaba u katunil” is here translated “it was the eleventh ahau katun,” while in paragraph 3 a similar sentence is translated “4 Ahau was the name of the katun.”

Thus an ahau katun is not mentioned in the Books of Chilan Balam in spite of Brinton’s translation, and neither is it mentioned by Landa, in spite of Brasseur de Bourbourg’s insertions. In fact the only evidence of it is the

assertion of Don Pio Perez. I do not mean to say that the period of 6×52 years = 312 years may not have had a name, nor that “ahau katun” may not be the name of some period, but so far we have no trustworthy contemporary evidence of either of these statements.

Transcriber's Note

The cover image was created by the transcriber and is placed in the public domain.

*** END OF THE PROJECT GUTENBERG EBOOK MAYAN
NOMENCLATURE ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.